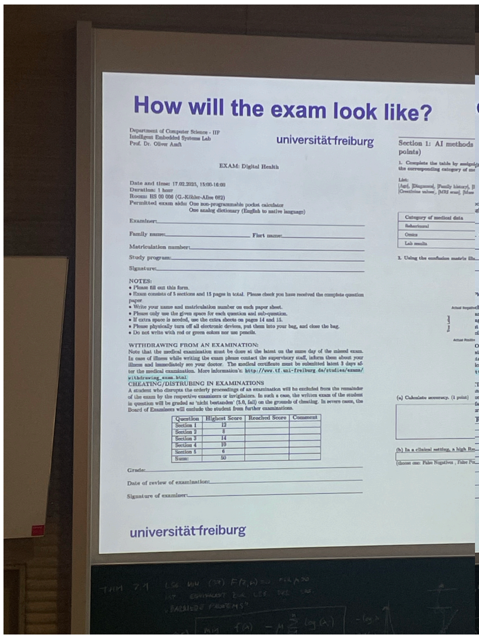
last years exam questions



How will the exam look like?

Department of Computer Science - IIF
Intelligent Embedded Systems Lab
Prof. Dr. Oliver Amft

universität freiburg

EXAM: Digital Health

Date and time: 17.02.2025, 15:00-16:00
Duration: 1 hour
Room: HS 00 006 (G.-Köhler-Allee 082)
Permitted exam aids: One non-programmable pocket calculator
One analog dictionary (English to native language)

Examiner: __________

Family name: __________ First name: __________

Matriculation number: __________

Study program: __________

Signature: __________

NOTES:
- Please fill out this form.
- Exam consists of 5 sections and 15 pages in total. Please check you have received the complete question paper.
- Write your name and matriculation number on each paper sheet.
- Please only use the given space for each question and sub-question.
- If extra space is needed, use the extra sheets on pages 14 and 15.
- Please physically turn off all electronic devices, put them into your bag, and close the bag.
- Do not write with red or green colors nor use pencils.

WITHDRAWING FROM AN EXAMINATION:
Note that the medical examination must be done at the latest on the same day of the missed exam. In case of illness while writing the exam please contact the supervisory staff, inform them about your illness and immediately see your doctor. The medical certificate must be submitted latest 3 days after the medical examination. More information's: http://www.tf.uni-freiburg.de/studies/exams/withdrawing_exam.html

CHEATING/DISTRUBING IN EXAMINATIONS
A student who disrupts the orderly proceedings of an examination will be excluded from the remainder of the exam by the respective examiners or invigilators. In such a case, the written exam of the student in question will be graded as 'nicht bestanden' (5.0, fail) on the grounds of cheating. In severe cases, the Board of Examiners will exclude the student from further examinations.

Question	Highest Score	Reached Score	Comment
Section 1	12		
Section 2	8		
Section 3	14		
Section 4	10		
Section 5	6		
Sum:	50		

Grade: __________

Date of review of examination: __________

Signature of examiner: __________

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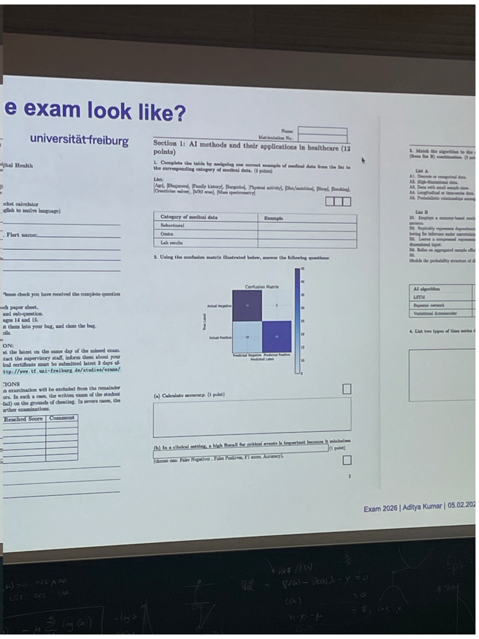


e exam look like?

Name: ☐
Matriculation No.: ☐

Section 1: AI methods and their applications in healthcare (12 points)

1. Complete the table by assigning one correct example of medical data from the list to the corresponding category of medical data. (3 points)

List:
[Age], [Diagnoses], [Family history], [Surgeries], [Physical activity], [Diet/nutrition], [Sleep], [Smoking], [Creatinine values], [MRI scan], [Mass spectrometry]

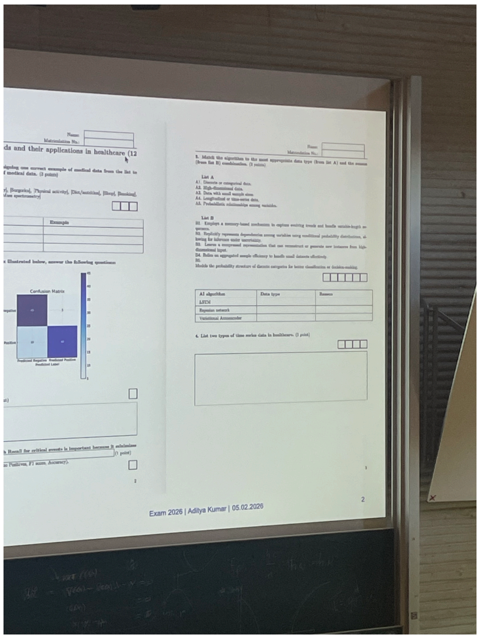
☐☐☐

Category of medical data	Example
Behavioural	
Omics	
Lab results	

2. Using the confusion matrix illustrated below, answer the following questions:

(a) Calculate accuracy. (1 point) ☐

(b) In a clinical setting, a high Recall for critical events is important because it minimizes __________ (1 point)
(choose one: False Negatives , False Positives, F1 score, Accuracy). ☐



Name: ☐
Matriculation No.: ☐

3. Match the algorithm to the most appropriate data type (from list A) and the reason (from list B) combination. (3 points)

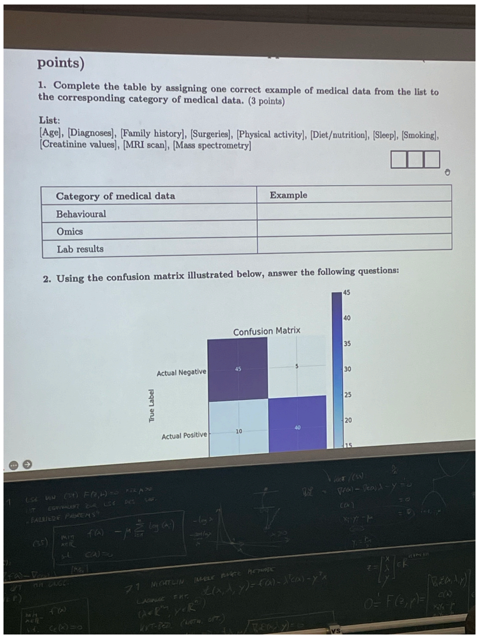
List A
A1. Discrete or categorical data.
A2. High-dimensional data.
A3. Data with small sample sizes.
A4. Longitudinal or time-series data.
A5. Probabilistic relationships among variables.

List B
B1. Employs a memory-based mechanism to capture evolving trends and handle variable-length sequences.
B2. Explicitly represents dependencies among variables using conditional probability distributions, allowing for inference under uncertainty.
B3. Learns a compressed representation that can reconstruct or generate new instances from high-dimensional input.
B4. Relies on aggregated sample efficiency to handle small datasets effectively.
B5.
Models the probability structure of discrete categories for better classification or decision-making.

☐☐☐☐☐☐

AI algorithm	Data type	Reason
LSTM		
Bayesian network		
Variational Autoencoder		

4. List two types of time series data in healthcare. (1 point) ☐☐☐☐



points)

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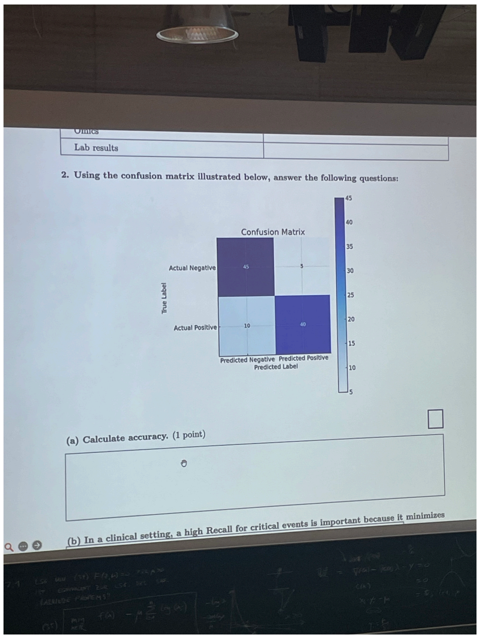
List:
[Age], [Diagnoses], [Family history], [Surgeries], [Physical activity], [Diet/nutrition], [Sleep], [Smoking], [Creatinine values], [MRI scan], [Mass spectrometry]

Category of medical data | Example
Behavioural
Omics
Lab results

2. Using the confusion matrix illustrated below, answer the following questions:

Confusion Matrix
Actual Negative
Actual Positive
True Label
45
5
10
40



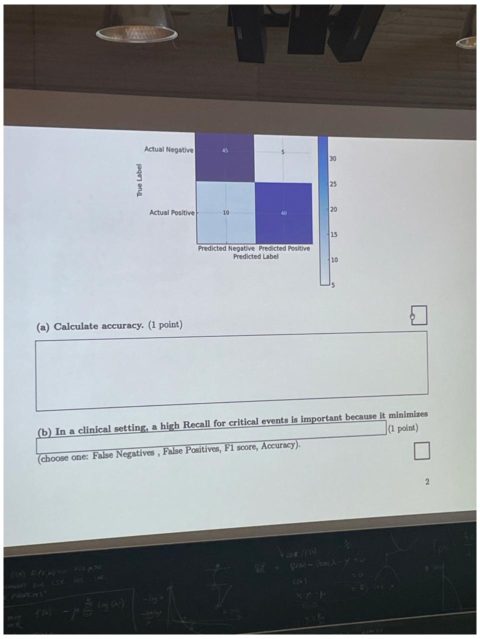
Omics
Lab results

2. Using the confusion matrix illustrated below, answer the following questions:

(a) Calculate accuracy. (1 point)

(b) In a clinical setting, a high Recall for critical events is important because it minimizes

Confusion Matrix
Actual Negative
Actual Positive
True Label
Predicted Negative
Predicted Positive
Predicted Label
45
5
10
40
45
40
35
30
25
20
15
10
5

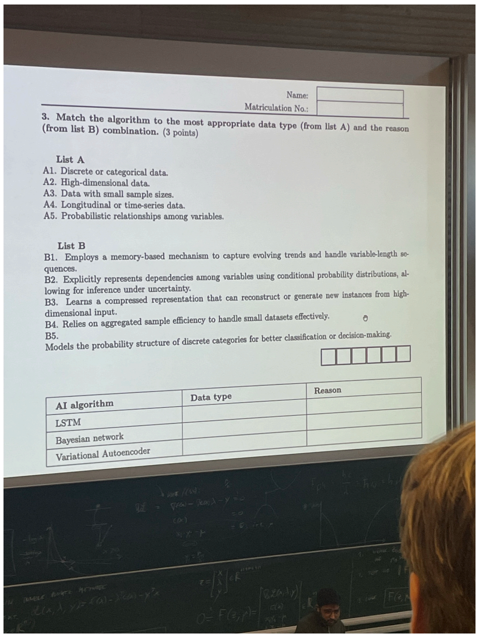


(a) Calculate accuracy. (1 point)

(b) In a clinical setting, a high Recall for critical events is important because it minimizes (1 point)
(choose one: False Negatives , False Positives, F1 score, Accuracy).

2

Actual Negative
Actual Positive
True Label
Predicted Negative
Predicted Positive
Predicted Label
45
5
10
40
30
25
20
15
10
5



Name:
Matriculation No.:

3. Match the algorithm to the most appropriate data type (from list A) and the reason (from list B) combination. (3 points)

List A
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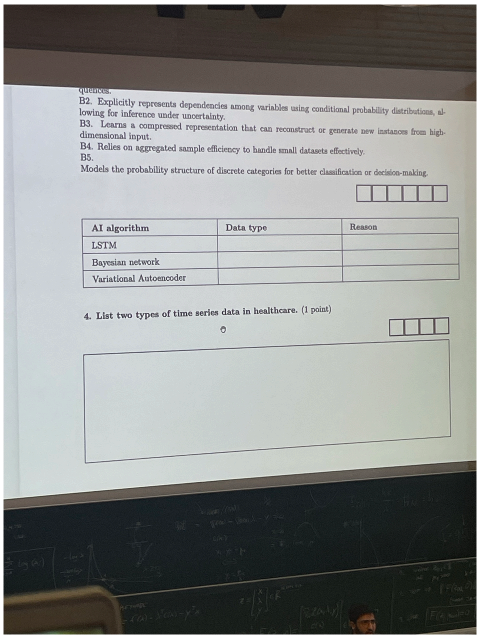
AI algorithm | Data type | Reason
LSTM
Bayesian network
Variational Autoencoder



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AI algorithm | Data type | Reason
LSTM
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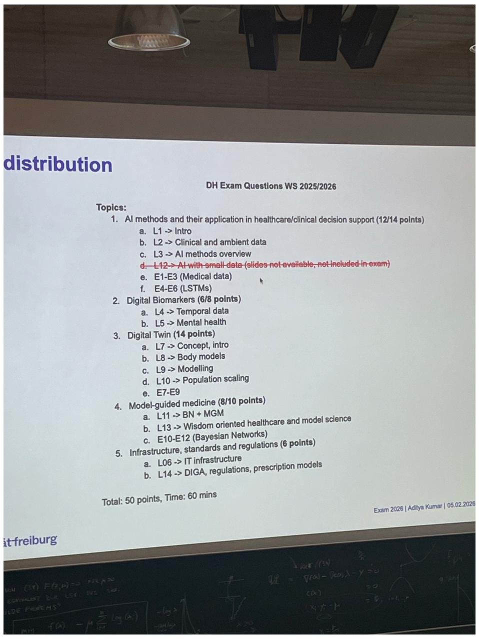
Variational Autoencoder

4. List two types of time series data in healthcare. (1 point)

Sections & what to learn



distribution

DH Exam Questions WS 2025/2026

Topics:

1. AI methods and their application in healthcare/clinical decision support (**12/14 points**)
    - a. L1 -> Intro
    - b. L2 -> Clinical and ambient data
    - c. L3 -> AI methods overview
    - ~~d. L12 -> AI with small data (slides not available, not included in exam)~~
    - e. E1-E3 (Medical data)
    - f. E4-E6 (LSTMs)
2. Digital Biomarkers (**6/8 points**)
    - a. L4 -> Temporal data
    - b. L5 -> Mental health
3. Digital Twin (**14 points**)
    - a. L7 -> Concept, intro
    - b. L8 -> Body models
    - c. L9 -> Modelling
    - d. L10 -> Population scaling
    - e. E7-E9
4. Model-guided medicine (**8/10 points**)
    - a. L11 -> BN + MGM
    - b. L13 -> Wisdom oriented healthcare and model science
    - c. E10-E12 (Bayesian Networks)
5. Infrastructure, standards and regulations (**6 points**)
    - a. L06 -> IT infrastructure
    - b. L14 -> DIGA, regulations, prescription models

Total: 50 points, Time: 60 mins

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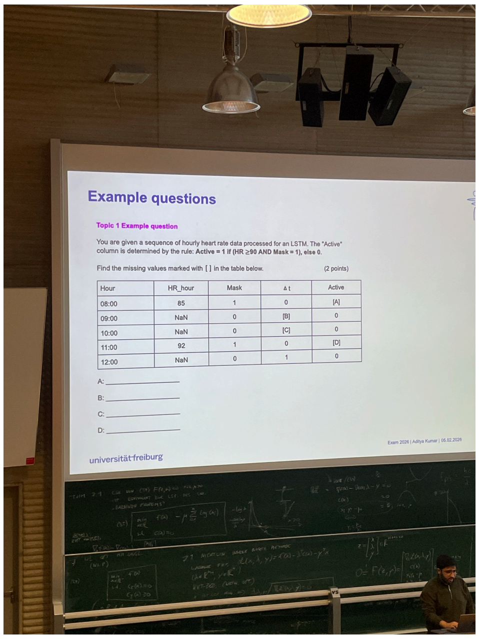
Example question 1



Example questions

Topic 1 Example question

You are given a sequence of hourly heart rate data processed for an LSTM. The "Active" column is determined by the rule: **Active = 1 if (HR ≥90 AND Mask = 1), else 0.**

Find the missing values marked with [ ] in the table below. (2 points)

Hour	HR_hour	Mask	Δt	Active
08:00	85	1	0	[A]
09:00	NaN	0	[B]	0
10:00	NaN	0	[C]	0
11:00	92	1	0	[D]
12:00	NaN	0	1	0

A: ____________

B: ____________

C: ____________

D: ____________

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-> solution:

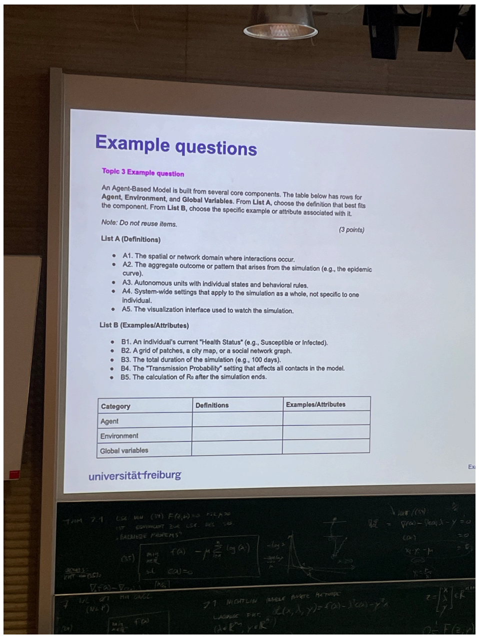
A: 0

B: 1

C: 2

D: 1

Example question 2



Example questions

Topic 3 Example question

An Agent-Based Model is built from several core components. The table below has rows for **Agent, Environment,** and **Global Variables.** From **List A,** choose the definition that best fits the component. From **List B,** choose the specific example or attribute associated with it.

Note: Do not reuse items. *(3 points)*

List A (Definitions)

- A1. The spatial or network domain where interactions occur.
- A2. The aggregate outcome or pattern that arises from the simulation (e.g., the epidemic curve).
- A3. Autonomous units with individual states and behavioral rules.
- A4. System-wide settings that apply to the simulation as a whole, not specific to one individual.
- A5. The visualization interface used to watch the simulation.

List B (Examples/Attributes)

- B1. An individual's current "Health Status" (e.g., Susceptible or Infected).
- B2. A grid of patches, a city map, or a social network graph.
- B3. The total duration of the simulation (e.g., 100 days).
- B4. The "Transmission Probability" setting that affects all contacts in the model.
- B5. The calculation of R_0 after the simulation ends.

Category	Definitions	Examples/Attributes
Agent		
Environment		
Global variables		

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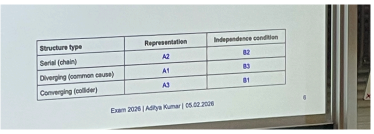
->

Solution:



Structure type	Representation	Independence condition
Serial (chain)	A2	B2
Diverging (common cause)	A1	B3
Converging (collider)	A3	B1

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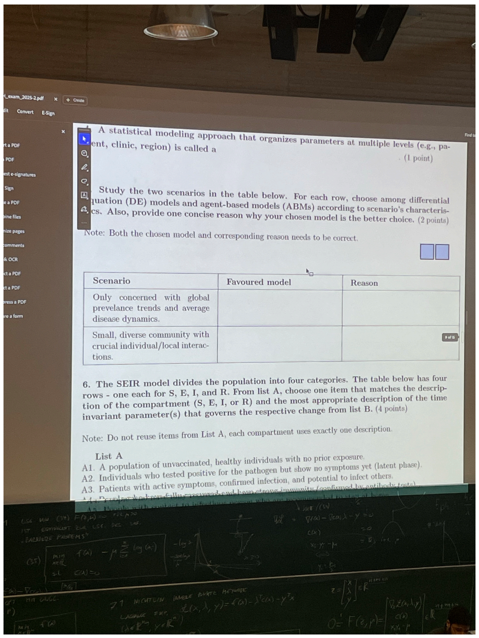
6

Here is a question, where you have to choose between given answers, but also think in your own words about reasons ->



A statistical modeling approach that organizes parameters at multiple levels (e.g., pa- ent, clinic, region) is called a

(1 point)

Study the two scenarios in the table below. For each row, choose among differential quation (DE) models and agent-based models (ABMs) according to scenario's characteris- cs. Also, provide one concise reason why your chosen model is the better choice. (2 points)

Note: Both the chosen model and corresponding reason needs to be correct.

Scenario	Favoured model	Reason
Only concerned with global prevelance trends and average disease dynamics.		
Small, diverse community with crucial individual/local interactions.		

6. The SEIR model divides the population into four categories. The table below has four rows - one each for S, E, I, and R. From list A, choose one item that matches the description of the compartment (S, E, I, or R) and the most appropriate description of the time invariant parameter(s) that governs the respective change from list B. (4 points)

Note: Do not reuse items from List A, each compartment uses exactly one description.

List A

A1. A population of unvaccinated, healthy individuals with no prior exposure.

A2. Individuals who tested positive for the pathogen but show no symptoms yet (latent phase).

A3. Patients with active symptoms, confirmed infection, and potential to infect others.

A4. People who have fully recovered and have strong immunity (confirmed by antibody tests).